Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
F		**2600043**

Summary

Use this template to begin typing the first page (summary page) of your electronic report. This template uses a 12-point Times New Roman font. Submit your paper as an Adobe PDF electronic file (e.g. 1111111.pdf), typed in English, with a readable font of at least 12-point type.

Do not include the name of your school, advisor, or team members on this or any page.

Be sure to change the control number and problem choice above.

You may delete these instructions as you begin to type your report here.

Follow us @COMAPMath on Twitter or COMAPCHINAOFFICIAL on Weibo for the most up to date contest information.

Keywords: keyword1; keyword2

Contents

1 Introduction

1.1 Problem Background

In recent years, generative artificial intelligence has become deeply integrated into learning and work, supporting writing, programming, design, information retrieval, and content creation while reshaping workflows in many roles. Employer surveys indicate that demand for key skills will shift significantly in the coming years. Skills gaps have become a major challenge for companies, and many now regard employee reskilling and upskilling as a central strategy.

Different professions are affected in different ways. In some occupations, a large share of tasks can be supported or partially undertaken by generative AI, while others rely more on on-site operations or interpersonal collaboration. Task-level estimates suggest that the jobs of roughly one quarter of workers worldwide will be affected to some degree, but the more common outcome is the reorganization and upgrading of work rather than the direct replacement of positions. This creates urgent pressure on education systems to adjust curricula and assessment methods so that students learn to use generative AI to enhance their capabilities while limiting negative impacts, at a time when many countries still lack adequate regulation and data privacy protection and educational institutions have not yet formed reliable practices for integrating these tools into teaching and research.

At the same time, the spread of generative AI brings resource and environmental pressures, as model training and deployment consume large amounts of electricity and hardware cooling increases water use, affecting local water supplies and ecosystems. Against this backdrop, this study focuses on linking changes in the world of work with educational decision-making through a data-driven approach. We analyze task changes and evolving skill demands across different occupations, then translate these findings into actionable adjustment plans for schools and programs, covering curriculum content, practical training, assessment design, faculty allocation, and risk management such as usage guidelines and rules for attribution and acknowledgment. Through multi-metric evaluation and uncertainty analysis, we aim to provide more robust recommendations and to build reusable decision-making frameworks for a wide range of training programs.

1.2 Restatement of the Problem

Problem 1: Design a clear, data-driven method that, against the backdrop of Gen-AI-induced changes in job tasks and skill demands, first screens and organizes a large set of occupations, and then selects a small number of representative occupations for more in-depth, targeted studies at the institutional level. We begin by using an occupationyear panel to build a composite index covering six dimensions: labor demand, income returns, skill structure, technology use, adaptability, and institutional environment. We then clean and standardize the data and apply an entropy-weight method to obtain annual composite scores for each occupation. Next, we use the K-means algorithm to cluster occupations based on recent score trends and identify several core occupation types. Finally, we forecast the development trends of each type and select one target occupation each from STEM, trades, and arts for subsequent analysis.

Placeholder for a chart summarizing projected skill-demand shifts and the share of firms prioritizing reskilling/upskilling.

Figure 1: Projected shifts in key skill demand and employer emphasis on reskilling/upskilling (survey-based).

Placeholder for a chart showing task-level exposure to generative AI across occupations (e.g., histogram/stacked bars by occupation group).

Figure 2: Estimated distribution of task-level exposure to generative AI across occupations, highlighting heterogeneous impacts.

1.3 Literature Review

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

- the angular velocity of the bat,
- the velocity of the ball, and
- the position of impact along the bat.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula. *center of percussion* [Brody 1986], Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.

Theorem 1.1. *LaTeX*

Lemma 1.2. *TeX.*

Proof. The proof of theorem. □

1.4 Our Work

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

-
-
-
-

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

2 Assumptions and Justifications

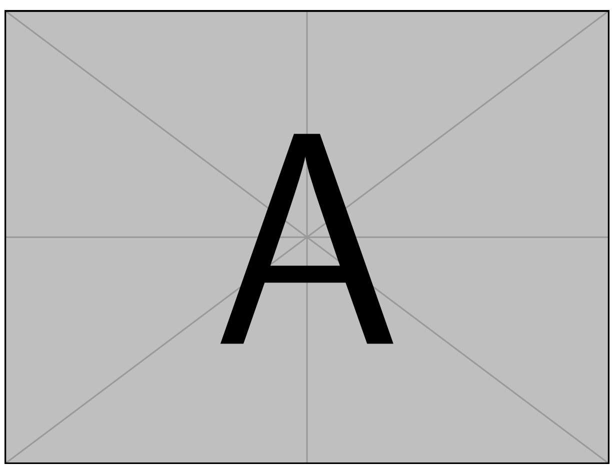

Figure 3: The name of figure

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Donec odio elit, dictum in, hendrerit sit amet, egestas sed, leo. Praesent feugiat sapien aliquet odio. Integer vitae justo. Aliquam vestibulum fringilla lorem. Sed neque lectus, consectetuer at, consectetuer sed, eleifend ac, lectus. Nulla facilisi. Pellentesque eget lectus. Proin eu metus.

Sed porttitor. In hac habitasse platea dictumst. Suspendisse eu lectus. Ut mi mi, lacinia sit amet, placerat et, mollis vitae, dui. Sed ante tellus, tristique ut, iaculis eu, malesuada ac, dui. Mauris nibh leo, facilisis non, adipiscing quis, ultrices a, dui. (1)

$$a^2 \tag{1}$$

$$\begin{pmatrix} *20ca_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \frac{Opposite}{Hypotenuse} \cos^{-1}\theta \arcsin\theta$$

Morbi luctus, wisi viverra faucibus pretium, nibh est placerat odio, nec commodo wisi enim eget quam. Quisque libero justo, consectetuer a, feugiat vitae, porttitor eu, libero. Suspendisse sed mauris vitae elit sollicitudin malesuada. Maecenas ultricies eros sit amet ante. Ut venenatis velit. Maecenas sed mi eget dui varius euismod. Phasellus aliquet volutpat odio. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque sit amet pede ac sem eleifend consectetuer. Nullam elementum, urna vel imperdiet sodales, elit ipsum pharetra ligula, ac pretium ante justo a nulla. Curabitur tristique arcu eu metus. Vestibulum lectus. Proin mauris. Proin eu nunc eu urna hendrerit faucibus. Aliquam auctor, pede consequat laoreet varius, eros tellus scelerisque quam, pellentesque hendrerit ipsum dolor sed augue. Nulla nec lacus.

$$p_j = \begin{cases} 0, & \text{if } j \text{ is odd} \\ r!\,(-1)^{j/2}, & \text{if } j \text{ is even} \end{cases}$$

Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui. Morbi ultrices rutrum lorem. Nam elementum ullamcorper leo. Morbi dui. Aliquam sagittis. Nunc placerat. Pellentesque tristique sodales est. Maecenas imperdiet lacinia velit. Cras non urna. Morbi eros pede, suscipit ac, varius vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetuer odio sem sed wisi.

$$\arcsin\theta = \iiint_{\varphi} \lim_{x\to\infty} \frac{n!}{r!\,(n-r)!} \tag{1}$$

3 Data Sources and Preprocessing

4 Notations

For each indicator X, the subscript i denotes the i-th career. The superscript (t) denotes the year index. i.e. $X_i^{(2018)}$ is the value of that indicator for career i in 2018.

Table 1: Indicator notations for the six dimensions.

Symbol	Indicator	Meaning	Unit
		Dim1: Labor Market Demand (LMD)	
$\mathrm{LMD1}_i^{(t)}$	Employment size	Number of employed persons in career i in year t	persons
$\mathrm{LMD2}_i^{(t)}$	Employment growth rate	Year-over-year employment growth rate of career i at year t	%
$\mathrm{LMD3}_i^{(t)}$	Vacancy rate	Job openings rate of career i at year t	%
$\mathrm{LMD4}_i^{(t)}$	Unemployment rate	Unemployment rate associated with career i at year t	%
$\mathrm{LMD5}_i^{(t)}$	New job creation rate	Rate of newly created jobs for career i at year t	%
		Dim2: Compensation and Economic Return (CER)	
$\mathrm{CER1}_i^{(t)}$	Median annual wage	Income level	USD/year
$\mathrm{CER2}_i^{(t)}$	Wage growth rate	Return growth	%
$\mathrm{CER3}_i^{(t)}$	Income dispersion	Within-occupation wage inequality	index
$\mathrm{CER4}_i^{(t)}$	Education return ratio	Income per training/education cost	ratio
$\mathrm{CER5}_i^{(t)}$	Non-wage benefits index	Stability and benefits	index
		Dim3: Skill Structure (SKS)	
$\mathrm{SKS1}_i^{(t)}$	Share of high-skill positions	Cognitive/professional skill intensity	%
$\mathrm{SKS2}_i^{(t)}$	Skill update frequency	Speed of skill iteration	1/year
$\mathrm{SKS3}_i^{(t)}$	Educational requirement threshold	Level of formal education required	level
$\mathrm{SKS4}_i^{(t)}$	Cross-disciplinary skill demand	Need for hybrid/combined competencies	index
$\mathrm{SKS5}_i^{(t)}$	On-the-job training intensity	Lifelong learning requirement	hours/year
		Dim4: Technology Dependence (TED)	
$\mathrm{TED1}_i^{(t)}$	Digital tool usage rate	Degree of technology penetration in daily tasks	%
$\mathrm{TED2}_i^{(t)}$	Automation potential index	Substitutability of tasks by automation	index
$\mathrm{TED3}_i^{(t)}$	Technology investment intensity	IT / capital investment intensity	USD/year
$\mathrm{TED4}_i^{(t)}$	Workflow standardization level	Ease of systematization and process codification	index
$\mathrm{TED5}_i^{(t)}$	Human–machine collaboration share	Extent of technology-enabled collaboration	%
		Dim5: Innovation and Adaptability (INA)	

Table 2: Core notations used in Sub-Model I.

Symbol	Descriptions	Unit
N	The number of careers (occupations)	count
m	The number of indicators	count
n	The number of years in the panel window	year
K	The number of clusters (archetypes)	count
L	Look-back length used to form clustering features (recent years)	year
H	Forecast horizon for archetype trend projection	year
$x_{ij}^{(t)}$	Raw indicator value of career i on indicator j in year t	indicator-dependent
$z_{ij}^{(t)}$	Normalized indicator after direction harmonization and min–max scaling	–
w_j	Entropy-derived indicator weight	–
$\mathrm{score}_i^{(t)}$	Annual composite career score, $\sum_{j=1}^{m} w_j z_{ij}^{(t)}$	–
$\mathbf{u}_i$	Recent-score feature vector for clustering, $\mathbf{u}_i \in \mathbb{R}^L$	–
$\mathcal{C}_k$	Set of careers assigned to archetype (cluster) k	–
$\boldsymbol{\mu}_k$	Cluster centroid in the feature space	–
$\overline{\mathrm{score}}_k^{(t)}$	Archetype mean score (cluster mean) in year t	–

5 Multi-Objective Optimization Model

5.1 Sub-Model I: Entropy-Weight-Based Career Screening and Representative Occupation Identification via K-Means

5.1.1 Indicator Selection and Data Preparation

Sub-Model I first develops a unified indicator system to describe how different careers have evolved over the past n years in the context of the ongoing development of generative AI (Gen-AI). Let $i = 1, \dots, N$ index careers, $j = 1, \dots, m$ index indicators, and $t \in t_1, \dots, t_n$ index years. For each careeryear observation, the study compiles a vector of raw indicator values $x_{ij}^{(t)}$ spanning six dimensions: labor market demand, economic return, skill structure, dependence on technology, innovation and adaptability, and the institutional environment.

Data Processing. To ensure comparability across datasets, we first discard indicators with excessive missing values. For the remaining missing entries, we impute them using withingroup summary statistics based on career family and year. We then harmonize the direction of all indicators so that larger values indicate better performance, and apply minmax normalization to rescale them to the interval $[0, 1]$, obtaining $z_{ij}^{(t)}$. This prevents indicators with inherently large magnitudes, such as wages, from dominating the composite score.

Table 3: Indicator system for Sub-Model I (with units and direction). (To be inserted)

5.1.2 Entropy Weighting and K-Means Clustering

After normalization, the analysis proceeds with two core objectives. First, multi-dimensional indicators are aggregated into a single annual score so that observations remain comparable across both occupations and years. Second, a small set of occupational archetypes is identified that is statistically coherent and conceptually interpretable, thereby supporting subsequent decision-making.

To avoid subjective weighting, this study adopts the entropy weight method (EWM). EWM assigns larger weights to indicators that exhibit greater dispersion in the occupation–year panel and therefore contain more discriminative information. Conversely, indicators that remain relatively stable across occupations provide limited screening power and receive lower weights through the entropy mechanism. Let w_j denote the entropy-based weight of indicator j, computed from the full panel. The annual composite score of occupation i in year t is defined as

$$\mathrm{score}_i^{(t)} = \sum_{j=1}^{m} w_j \, z_{ij}^{(t)}. \tag{2}$$

By aggregating the annual scores, we obtain the multiyear composite score trajectory for career i:

$$\mathrm{score}_i^{(t_1 \sim t_n)} = \left[\mathrm{score}_i^{(t_1)}, , \mathrm{score}_i^{(t_2)}, , \ldots, , \mathrm{score}_i^{(t_n)}\right]. \tag{3}$$

This trajectory reflects both the overall level of career development and its evolution over time. The weights obtained in this calculation are shown in Figure 4. They not only provide an intuitive view of which indicator dimensions are the primary drivers of crosscareer differences in the sample, but also offer an interpretable basis for the subsequent clustering and forecasting analyses.

Figure 4: Entropy-derived indicator weights w_j (higher values indicate stronger discriminatory power). (To be inserted)

Careers are clustered based on the similarity of their recent score trajectories in order to identify typical occupational archetypes. Specifically, each career i is transformed into a compact feature vector composed of its most recent L annual scores:

$$\mathbf{u}_i = \left[\mathrm{score}_i^{(t_{n-L+1})}, \ldots, \mathrm{score}_i^{(t_n)}\right] \in \mathbb{R}^L, \tag{4}$$

so that careers with similar development levels and trends lie closer to one another in this L-dimensional space.

K-means clustering is then applied to the set of feature vectors $\{\mathbf{u}_i\}$, partitioning them into K groups by minimizing within-cluster dispersion and thereby producing directly usable labels for career archetypes:

$$\min_{\{\mathcal{C}_k\}_{k=1}^K} \sum_{k=1}^{K} \sum_{\mathbf{u}_i \in \mathcal{C}_k} \left\| \mathbf{u}_i - \boldsymbol{\mu}_k \right\|_2^2 , \tag{5}$$

where $\mathcal{C}_k$ and $\boldsymbol{\mu}_k$ denote cluster k and its centroid, respectively. The number of clusters K is selected using elbow and silhouette diagnostics to strike a balance between parsimony, meaning fewer and more interpretable archetypes, and fidelity, meaning sufficient separation among distinct score-trajectory patterns.

Finally, clustering results are visualized using a two-dimensional t-SNE embedding (Figure 5). This embedding is used solely for visualization and does not affect the K-means solution.

Figure 5: Two-dimensional t-SNE visualization of careers colored by K-means cluster labels. (To be inserted)

5.1.3 Results: Archetype Interpretation, Representative Careers, and Trend Projection

This subsection turns the clustering output into results that directly serve the research objectives. Because the problem statement already specifies three broad category labelsSTEM, Trade, and Artswe first examine whether the clustering solution is reasonably consistent with this taxonomy. For several candidate values of the number of clusters K, we compute the Adjusted Rand Index and the silhouette coefficient as a measure of between-cluster separation, and then select the final K by balancing interpretability against empirical clustering quality. We emphasize that a moderate Adjusted Rand Index is considered the desirable outcome, as it indicates that the extracted occupational archetypes capture structural patterns that are related to, but not merely a simple replica of, the original labels.

Table 4: Model selection summary for K: ARI (vs. STEM/Trade/Arts), silhouette, and notes. (To be inserted)

K	ARI	Silhouette	Notes

Once K is selected, each cluster is interpreted by combining its centroid-level indicator profile with its score trajectory relative to the overall distribution. In practice, the cluster mean trajectory $\overline{\text{score}}_k^{(t)}$ is used to describe the baseline level and recent direction of change, while the strongest contributing dimensions are used to diagnose whether the archetype is primarily characterized by demand-side strength, return stability, skill structure, or technology dependence. This interpretation step turns the statistical groupings into decision-relevant narratives, clarifying for each archetype whether it appears more vulnerable to task substitution or more likely to benefit from Gen-AI-enabled augmentation.

Finally, Sub-Model I projects medium-term trends at the level of occupational archetypes, which is more stable and less error-prone than forecasting each occupation individually. For each cluster k, the archetype score series is defined as the cluster mean:

$$\overline{\text{score}}_k^{(t)} = \frac{1}{|\mathcal{C}_k|} \sum_{i \in \mathcal{C}_k} \text{score}_i^{(t)}. \tag{6}$$

In view of the potential trend shift induced by the emergence of generative AI (with technical details omitted here), a parsimonious segmented trend model is fitted to this series, and bootstrap

Table 5: Career archetypes: interpretation summary, cluster size, and representative occupation. (To be inserted)

Cluster	Interpretation (key traits and Gen-AI exposure)	Size	Representative occupation
C1	[TBD]		[TBD]
C2	[TBD]		[TBD]
C3	[TBD]		[TBD]

methods are used to construct uncertainty bands for forecasts over the next H years. The resulting projections, combined with the interpretation of each occupational archetype, provide a clear and traceable basis for selecting three target careers for in-depth, institution-specific analysis, with exactly one career chosen from each of the STEM, trade, and arts categories.

Figure 6: Forecasted archetype score trajectories with uncertainty bands. (To be inserted)

5.2 Sub-Model II:

5.3 Sub-Model III:

6 Results and Extension

6.1 Optimization Objectives

6.2 Constraints Analysis

6.3 Optimization Result Analysis

6.4 Key Parameter Determination

7 Sensitivity Analysis

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

8 Strengths and Weaknesses

8.1 Strengths

8.2 Weaknesses

9 Conclusion

Appendices

Appendix A First appendix

In addition, your report must include a letter to the Chief Financial Officer (CFO) of the Goodgrant Foundation, Mr. Alpha Chiang, that describes the optimal investment strategy, your modeling approach and major results, and a brief discussion of your proposed concept of a return-on-investment (ROI). This letter should be no more than two pages in length.

Dear, Mr. Alpha Chiang

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Sincerely yours,

Your friends

Here are simulation programmes we used in our model as follow.

Input matlab source: *Source file ./code/mcmthesis-matlab1.m not found.*

Appendix B Second appendix

some more text **Input C++ source:** *Source file ./code/mcmthesis-sudoku.cpp not found.*

Report on Use of AI

1. Bing AI
Query1: write a sort algorithm
Output: Sure, I can help you with that. Here is a simple implementation of the selection sort algorithm in Python:

```
def selection_sort(arr):
    for i in range(len(arr)):
        min_idx = i
        for j in range(i + 1, len(arr)):
            if arr[min_idx] > arr[j]:
                min_idx = j
        arr[i], arr[min_idx] = arr[min_idx], arr[i]
    return arr
```

This algorithm sorts an array by repeatedly finding the minimum element from the unsorted part of the array and putting it at the beginning. The time complexity of this algorithm is O(n^2).

There are many other sorting algorithms available such as bubble sort, insertion sort, merge sort, quick sort, heap sort, and more. Each of these algorithms has its own advantages and disadvantages. You can learn more about them on GeeksforGeeks and Programiz.

I hope this helps!

2. OpenAl Enie (Nov 5, 2023 version, Ernie 4.0)
Query1: <insert the exact wording of any subsequent input into the Al tool>
Output: <insert the complete output from the second query>

3. Github CoPilot (Feb 3, 2024 version)
Query1: <insert the exact wording you input into the Al tool>
Output: <insert the complete output fron the Al tool>

4. Google Bard (Feb 2,2024 version)
Query: <insert the exact wording of your query>
Output: <irsert the complete output from the Al tool>

5. Bing AI
Query: write a FFT algorithm
Output: Sure, here is a Python implementation of the Fast Fourier Transform (FFT) algorithm:

```
import numpy as np

def fft(x):
    n = len(x)
    if n == 1:
        return x
    even = fft(x[0::2])
    odd = fft(x[1::2])
    T = [np.exp(-2j * np.pi * k / n) * odd[k] for k in range(n // 2)]
    return [even[k] + T[k] for k in range(n // 2)] + [even[k] - T[k] for k
       ↪ in range(n // 2)]
```

The FFT is a widely used algorithm for computing the **Discrete Fourier Transform (DFT)** of a sequence. It reduces the number of computations needed from $O(n^2)$ to $O(n \log n)$ 12. The basic idea of the FFT is to apply divide and conquer. We divide the coefficient vector of the polynomial into two vectors, recursively compute the DFT for each of them, and combine the results to compute the DFT of the complete polynomial 3.

I hope this helps!